Example 7.9

The length, breadth and height of a cuboid is 120 *mm*, 10 *cm* and 8 *cm* respectively. Find the volume of 10 such cuboids.

Solution

Since both breadth and height are given in *cm*, it is necessary to convert the length also in *cm*.

So we get, $l = 120\ mm = \frac{120}{10} = 12\ cm$ and take $b = 10\ cm$, $h = 8\ cm$ as such.

Volume of a cuboid $= l \times b \times h$

$= 12 \times 10 \times 8$

$= 960\ cm^3$

Volume of 10 such cuboids $= 10 \times 960$

$= 9600\ cm^3$

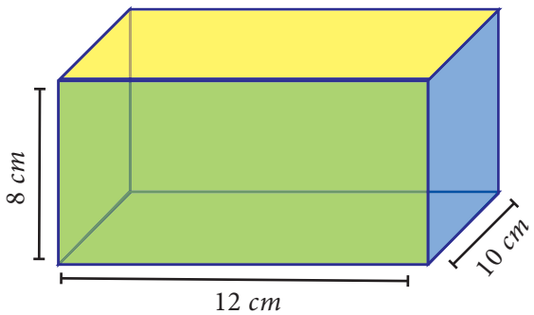


Fig. 7.20

Example 7.10

The length, breadth and height of a cuboid are in the ratio 7:5:2. Its volume is 35840 cm^3. Find its dimensions.

Solution

Let the dimensions of the cuboid be

$l = 7x,\ b = 5x$ and $h = 2x$.

Given that volume of cuboid = 35840 cm^3

$$l \times b \times h = 35840$$

$$(7x)(5x)(2x) = 35840$$

$$70x^3 = 35840$$

$$x^3 = \frac{35840}{70}$$

$$x^3 = 512$$

$$x = \sqrt[3]{8 \times 8 \times 8}$$

$$x = 8\ cm$$

Length of cuboid $= 7x = 7 \times 8 = 56\ cm$

Breadth of cuboid $= 5x = 5 \times 8 = 40\ cm$

Height of cuboid $= 2x = 2 \times 8 = 16\ cm$

Thinking Corner

Each cuboid given below has the same volume 120 cm^3. Can you find the missing dimensions?

(i)

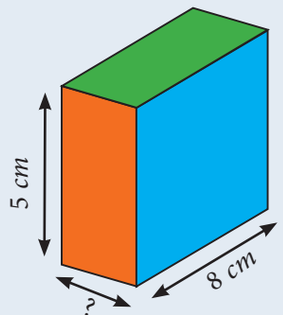


Fig. 7.21

(ii)

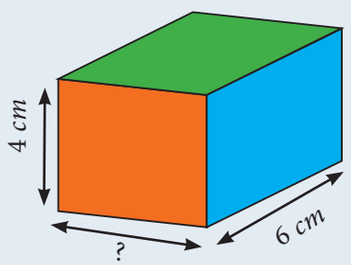


Fig. 7.22

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